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EEG-alpha rhythms and memory processes

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Abstract

The results of several experiments indicate that alpha frequency varies as a function of memory performance. It was found that in samples of age matched subjects alpha frequency of good memory performers is about 1 Hz higher than those of bad performers. The difference in alpha frequency between good and bad performers reaches a maximum during the retrieval of information, is much smaller during encoding and is minimal — but still significant — during a resing period. These results suggest that alpha frequency may be a permanent and not only a functional parameter that determines the speed with which information can be retrieved from memory. The calculation of changes in band power indicate further that the upper alpha band is particularly sensitive to semantic memory demands. The lower alpha band, on the other hand, seems to reflect attentional processes. These findings are discussed on the basis of a hypothesis which assumes that EEG frequencies within the alpha band stem at least in part from the thalamus and that the activity of thalamo–cortical networks reflects processes that are related to searching, accessing and retrieving information from (scmantic) long-term memory. © 1997 Elsevier Science B.V.

Keywords: Alpha frequency; Alpha power; Desynchronization; Memory attention

1. Introduction: memory search and the possible meaning of thalamo–cortical feedback loops

One of the crucial issues in memory research is based on the question of how a search process finds the relevant information in memory. Whereas most researchers would agree that (parts of) the cortical neural network can be considered a storage network for memory, little is known about the way a search process is initiated and how information is accessed and retrieved. Memory theories assume that a search process can be described as a spreading activation process in the cortical storage network (Klimesch, 1994). However, in order to find the relevant sought-after information, the critical assumption is that the activation status of the searched network must be transmitted back by means of feedback loops to a monitoring system. In the identification of the relevant codes that are parts of the searched network, feedback loops may play a decisive role. The basic idea is that a monitoring network converging in a particular monitoring system is mapped onto the storage network. Consequently, the monitoring system should be connected with the cortex by a dense network of axonal connections. Besides the basal ganglia, the thalamus with its thalamo–cortical projections to virtually all different cortical regions (Lopes da Silva et al., 1990; Höhl-Abrahao and Creutzfeldt, 1991) is one

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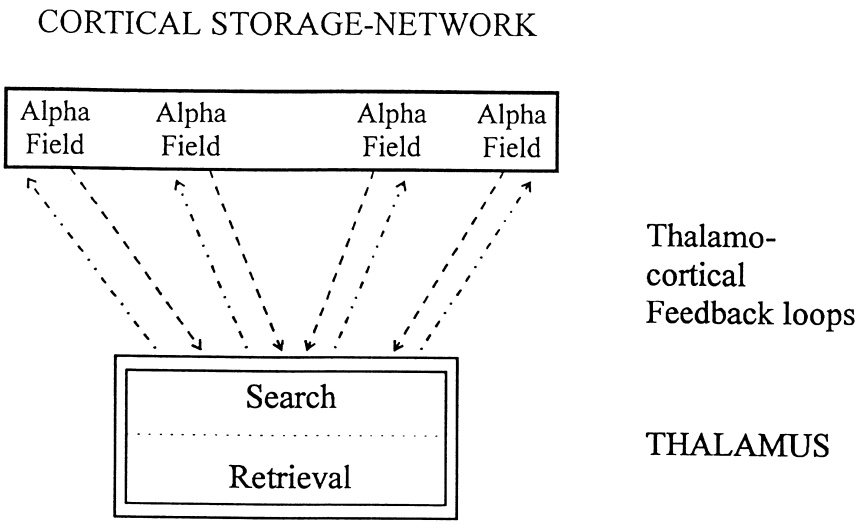


Fig. 1. A speculative view on the functional meaning of alpha frequency. Parts of the cortical neural network are considered a storage network for memory. A search process in memory spreads through the storage network automatically by means of self-organizing activation processes. However, the initiation of a search process and the retrieval of the relevant information is guided by a 'monitoring network' that is mapped onto the storage network. Thalamo–cortical feedback loops that are part of this network feed back the activation status of the cortical spreading activation process and thus allow to identify and to access the relevant sought-after information. It is assumed that an 'alpha field' and thalamo–cortical feedback loops oscillate within the alpha frequency range. Alpha desynchronization means 'activation', whereas synchronization reflects an inactive or resting state (see Pfurtscheller, 1992). Note that any cognitive process (such as thinking, perceiving, problem solving and speaking) depends on a fast access and retrieval of semantic information stored in long-term memory (LTM).

of those brain structures that fulfills this requirement. It is important to see that, as compared to the thalamo–cortical network, the cortical network is orders of magnitude denser. Thus, as schematically depicted in Fig. 1, each feedback loop serves a relatively large cortical field which also will be termed 'alpha field'.

Some researchers have suggested that the EEG frequency within the alpha band (of about 8–13 Hz) stems from the thalamus and induces synchronized neural activity in the cortex (Steriade et al., 1990; see also, e.g. the summary in Basar, 1992). If we proceed from the idea that memory codes are retrieved via longitudinal pathways linking thalamic nuclei with the cortex, and that alpha is the predominant rhythm reflecting the activity of these pathways, we arrive at the hypothesis that alpha frequency should be related to memory performance. We have tested this hypothesis (Klimesch et al., 1990, 1993) and found that as compared to bad memory performers, good performers show a significantly higher alpha frequency.

The starting point for an empirical evaluation of the hypothesis that alpha frequency is related to memory performance, is given by the following three assumptions: (1) Memory codes are represented by interconnected and widely distributed cell assemblies in the neocortex; (2) memory codes are accessed via longitudinal pathways (feedback loops) linking other brain areas such as the thalamus and the hippocampus with the neocortex; and (3) The desynchronization of alpha rhythms reflects the activity of some of those pathways and cortical areas which are involved in accessing, searching, and retrieving semantic long-term memory (LTM)-information. When focusing on alpha frequency, the basic prediction is that interindividual differences in alpha frequency reflect interindividual differences in memory performance.

2. Interindividual differences in alpha frequency and memory performance

In a first step, before testing this prediction in

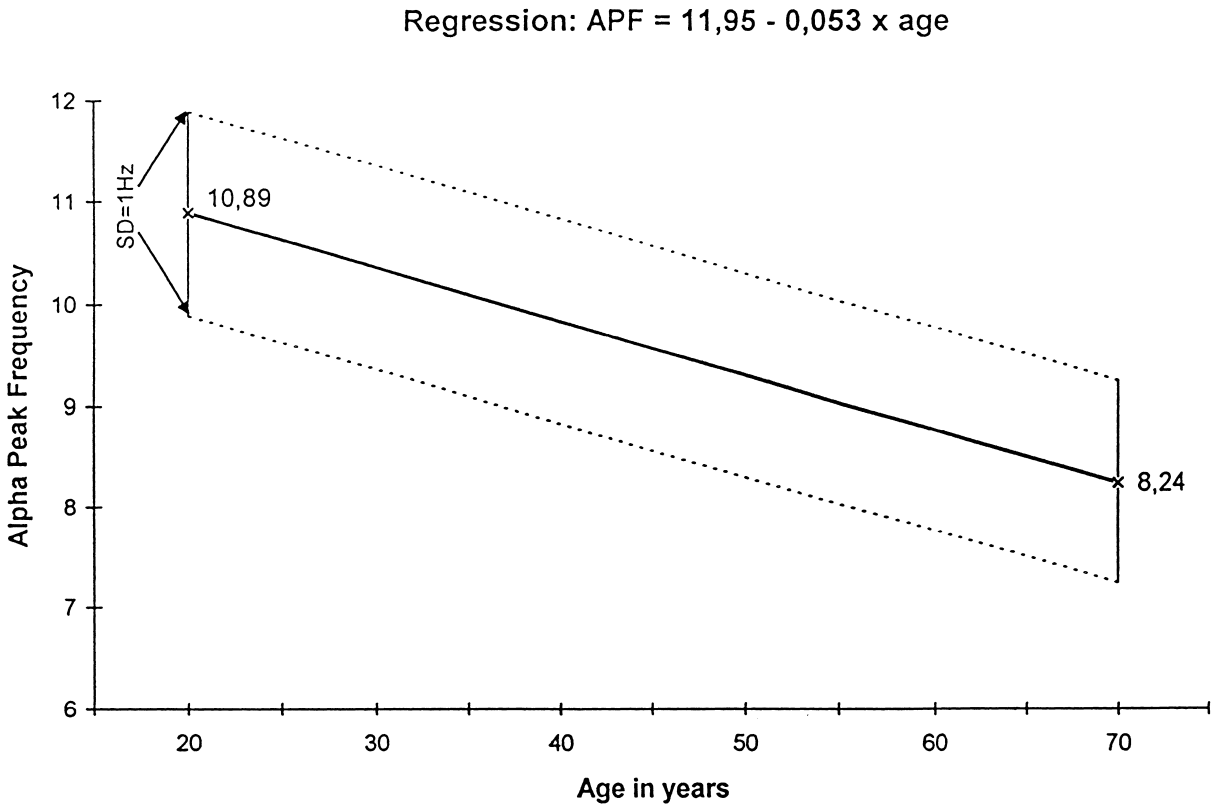


Fig. 2. Alpha peak frequency decreases with increasing age. Note that interindividual differences are of about the same magnitude as age dependent intraindividual differences within an age range of 20–70 years. Linear equation after Köpruner et al. (1984), values for S.D. from Klimesch et al. (1990) (1993).

detail, it is important to show whether alpha frequency does vary interindividually and what the factors are that affect alpha frequency. It is well known that alpha frequency decreases with increasing age. Köpruner et al. (1984) have found a linear relationship (alpha peak frequency = 11.95−0.053 × age) within the age range of adult subjects that is depicted in Fig. 2.

According to this relationship, a young adult of, e.g. 20 years has an expected peak frequency of 10.89 Hz, whereas a 70-year-old subject shows a drop of 2.65 Hz down to a frequency of 8.24 Hz. However, even subjects of the same age show a considerable variability in alpha frequency. In age controlled samples, alpha frequency shows a SD of about 1 Hz (Köpruner et al., 1984; Klimesch et al., 1990) which means that even for age matched subjects, an interindividual difference of about 2 Hz is quite a common case. Furthermore, clinical evidence demonstrates that a variety of brain diseases lowers alpha frequency (Köpruner et al., 1984). At the first glance this latter fact does not seem relevant with respect to our prediction because usually a general slowing of the overall EEG-frequency and not only of the alpha frequency is observed and because a variety of cognitive deficits may be correlated with a slowing of the EEG. There are, however, reports that indicate that a decrease in alpha activity might be selectively related to a decrease in memory performance. Sheridan et al. (1988) compared two groups of Alzheimer patients, one Group (Group A) showing regular alpha waves during eye closure (in the frequency range of 9–13 Hz) and attenuated by eye opening and a second Group (Group B) showing only irregular mixed frequency slow alpha (in the range of 7–11 Hz) without attenuation by eye opening. Most inter-

estingly, Sheridan et al. (1988) report that although both groups did not differ on the Mattis Demantia Scale, Group A had higher Wechsler Memory Scale scores (87 as compared to 70 in Group B) and higher parietal glucose metabolic rates. The reported differences were only marginally significant.

These preliminary results show that alpha frequency which usually is measured as peak frequency does vary to a considerable extent. However, this fact raises a methodological question. Because it has long been proposed (e.g. Grey Walter in Evans and Mulholland, 1969) and now is well accepted that there is no single alpha rhythm but instead a variety of different alpha rhythms, peak frequency might not be the best parameter to describe alpha frequency. If an entire population of alpha frequencies exist, the shape of the alpha power distribution will be more informative than peak frequency. Furthermore, in many cases the distribution of alpha power reveals multiple peaks and it appears quite arbitrary to select just the highest peak.

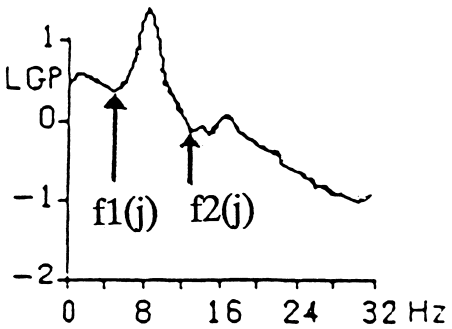
Center of gravity frequency is a measure which depends on the shape of the power distribution and, thus, is a measure that reflects the activity of the alpha population in a more adequate way. Another important reason to use gravity frequency is that peak frequency usually can be measured only if a subject is in rest. During perceptual or mental activity, alpha desynchronizes and the power distribution may become quite flat. In these cases, gravity frequency is the only reasonable choice for measuring alpha frequency. However, unlike peak frequency, gravity frequency varies with the selection of the exact frequency range within the alpha band. Using a fixed range (e.g. 8–13 Hz) when calculating gravity frequency does not make much sense. Due to interindividual differences in alpha frequency, for some subjects parts of the alpha power distribution will fall outside a fixed range. As a consequence, gravity frequency would be distorted and would not reflect interindividual differences in an adequate way. A promising way to avoid this problem is first to define the frequency range of the alpha power distribution individually for each electrode and each subject and then to calculate gravity frequency within that range (see Table 1). Because this procedure, which is described in Klimesch et al. (1990), results in an extended alpha band on the average, we use the term IAF (Individual Alpha Frequency) to avoid confusions with conventional measures.

Table 1
Steps for defining the bandwidth of individual alpha frequency (IAF)

(1) For a resting period, power spectra are calculated and plotted on a logarithmic scale for each subject and each of 29 recording sites.
(2) Then by visual inspection for each spectrum, the beginning of the 'ascent'($f1(i,j)$) and the end of the 'descent' ($f2(i,j)$) of the alpha peak are marked. i.....1, n = recording sites; j = ...1, k = subjects (3) The individual alpha bandwidth is given by the means $f1(j) = (f1(i,j))/n$ and $f2(j) = (f2(i,j))/n$ which are the averaged values over all recording sites per subject. (4) Within that frequency window, defined by $f1(j)$ and $f2(j)$, gravity frequency was calculated for each electrode and experimental condition. This frequency is termed IAF

In experiment 2, averaged over the entire sample, the mean values for determining IAF are: f1 = 7.6 Hz; f2 = 13.8 Hz.

Example:



For Experiment 2, averaged over the entire sample, the mean values for determining IAF are: f1 = 7.6 Hz, f2 = 13.8 Hz

IAF was used as indicator of alpha frequency in a series of experiments. In Experiment 1 of Klimesch et al. (1990), IAF was measured during a resting period in a sample of Alzheimer demented subjects (mean age = 71.5; SD = 7.76) who were also tested with the Wechsler Memory Scale.

Based on their overall quotients, subjects were divided in two groups, in a group with (comparatively) good memory (denoted M + with a mean age of 72 years) and in a group with bad memory (denoted M − with a mean age of 70 years). Comparing mean IAF between the two groups (M + : 8.14 Hz; M − : 7.02 Hz) shows that mean IAF differs in the predicted direction. Alzheimer subjects with good memory show an alpha frequency which is 1.12 Hz higher than Alzheimer subjects with bad memory. Correlations computed for individual IAF at all leads (the EEG was recorded from 19 electrodes) and the overall Wechsler scores over the entire sample revealed that at all recording sites of both hemispheres alpha frequency correlates significantly with memory performance as measured by the Wechsler Memory Scale. Thus, in agreement with the findings of Sheridan et al. (1988) our study provides additional support for a selective relationship between alpha frequency and memory performance. A weak, but not significant relationship between verbal memory and alpha frequency was reported by Saletu and Grünberger (1985). However, in contrast to our study and to Sheridan et al. (1988), Saletu and Grünberger (1985) used fixed frequency bands. From control computations with our data we know that fixed frequency bands tend to blur a significant relationship between IAF and memory performance. This is not surprising, because, as we have already emphasized, due to interindividual differences significant portions of the alpha power distribution fall outside a fixed frequency band, thus biasing the measurement of alpha frequency. The use of fixed frequency bands might very well have prevented other researches from detecting a relationship between alpha frequency and memory performance (e.g. Surwillo, 1971)

It should be noted that the results, reported so far, refer to alpha resting frequency. The interesting question, thus, is whether alpha frequency shifts from a reference interval to a test interval in which subjects actually are retrieving information from memory. In Experiment 2 of Klimesch et al. (1990) subjects performed a verbal recognition test and IAF was measured during a test interval of 1 s in which a subject was correctly recognizing a word and in a reference interval (of also 1 s) which preceded the test interval by 2 s. The results confirmed the findings of Experiment 1 and demonstrated again that, as compared to bad memory performers, good performers have an IAF which is more than 1 Hz higher. However, when comparing the reference with the test interval, neither an increase in IAF nor an increase in the significance of correlation coefficients between IAF and memory performance could be found.

3. Intraindividual differences in alpha frequency and memory performance

Although Experiments 1 and 2 support the hypothesis of a relationship between alpha frequency and memory performance, the results, reported so far, could theoretically also be explained in terms of interindividual differences in attention. It could be objected that the higher IAF of good memory performers reflects a state of increased attention which is responsible for an increase in memory performance. There is ample evidence that alpha desynchronization may reflect attention (Mulholland, 1969). This objection may appear particularly plausible within the context of Experiment 2, because IAF was measured during actual memory performance that of course also requires attentional resources. It is, however, less likely that resting IAF as measured in Experiment 1 also reflects attentional resources. For the resting period, subjects were asked to relax and interindividual differences in attention are an unlikely factor to determine differences in IAF. In support with this interpretation, Treisman (1984) has shown that intraindividual fluctuations in alpha frequency obviously are too small to account for fluctuations in attention. We assume that even during a resting state, the potential processing capacity of the thalamo–cortical network might very well be reflected by IAF.

In an attempt to rule out the possibility that IAF reflects attentional instead of memory processes, a specially designed experiment (denoted Experiment 3 in the following) was carried out by Klimesch et al. (1993). In this study, a modified version of Schneider's and Shiffrin's memory search paradigm (Schneider and Shiffrin,

1977) was used which allows to vary attentional and memory demands in an orthogonal design. In each trial subjects viewed a string of five or 10 characters (termed 'memory set') which they are asked to retain in memory. Then, after an interstimulus interval of 2 s, a single character (letter or number) appeared (termed frame). If the frame is an element of the memory set, it is called a target, otherwise a distractor. Subjects were asked to respond with 'yes' to a target and with 'no' to a distractor. Attentional demands are varied by using either the same memory set across a series of trials (termed consistent mapping condition) or different memory sets with different characters (termed varied mapping condition). Under the consistent mapping condition subjects knew, which characters the memory set on the next trial will contain. Under the varied mapping condition each memory set contains new characters. Thus, when deciding whether a frame is a target or distractor, subjects must be careful not to confuse characters between the current and the previously

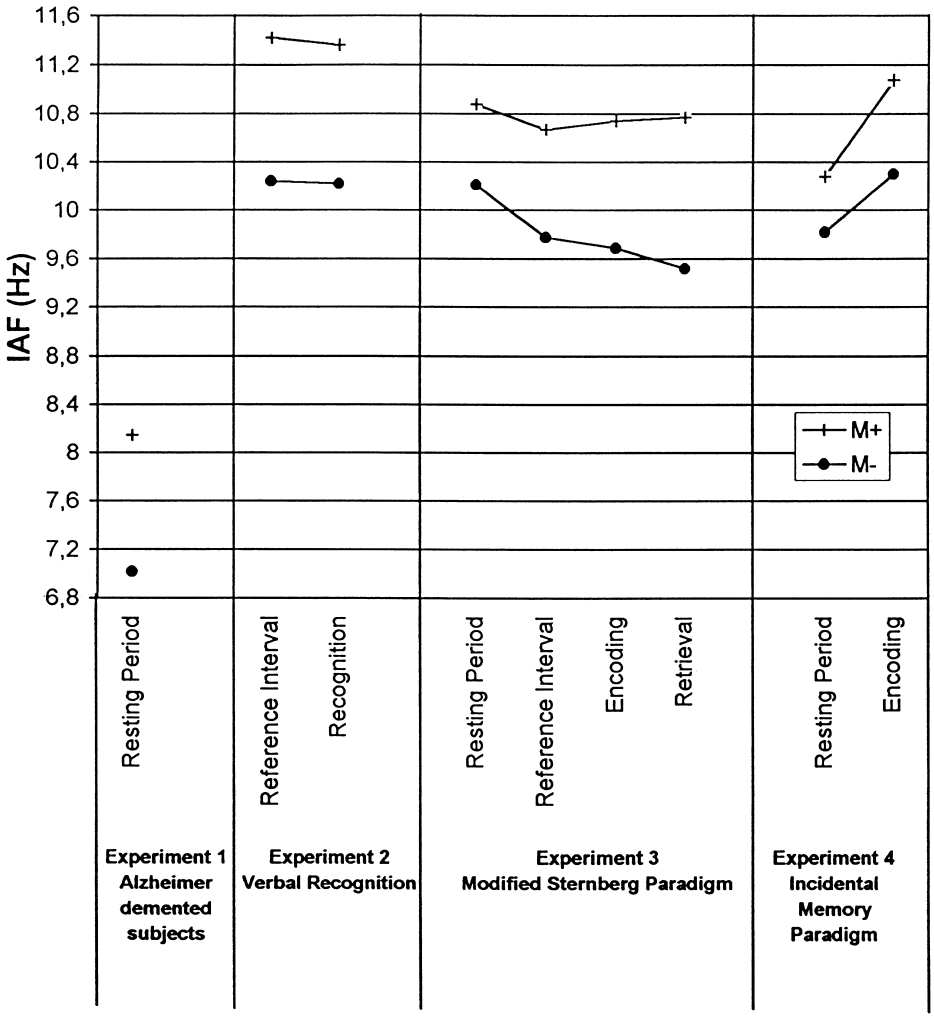


Fig. 3. Alpha frequency (IAF) for good memory performers (M +) is about 1 Hz higher as compared to subjects with comparatively bad memory performance (M −). This result was found in four different experiments. Note that the frequency difference between M + and M − increases as tasks demands increase.

shown memory set. According to Shiffrin and Schneider, selective (or focused) attention refers to those control mechanisms which allow a subject to encode a sensory input into memory. The concept of selective attention is closely linked to that of limited capacity. If processing capacity is overloaded, attention becomes divided and performance is reduced. As an example, if the memory set comprises only a single character, attention is 'focused' on the target. But with increasing memory set size, attention also becomes increasingly 'divided' and processing capacity overloaded. This design allows us to vary attentional demands (as manipulated by the two mapping conditions) and memory demands (as manipulated by memory set size) as independent factors while using selected groups of good and bad memory performers as subjects.

If attention would be the only relevant factor, IAF should vary as a function of attentional demands. If, on the other hand, memory processes are the only relevant factor, IAF should vary as a function of memory demands. Increased attention will result in increased memory performance. Because memory performance is reflected by IAF, we have to assume that both, attentional and memory demands vary with IAF. The crucial prediction, however, is that the manipulation of attentional demands does not blur or abolish the frequency difference between good (M +) and bad memory performers (M −). As the pronounced differences between good and bad memory performers show (see the right part of Fig. 3), this prediction is supported by the results of Experiment 3.

The variation of attentional demands affects IAF only in the range of 1/10th of 1 Hz as the difference between the varied and consistent mapping condition in Fig. 4 shows. Consistent effects in the sense that increasing attentional load leads to a decrease in IAF are found at central and parietal leads only.

Differences in memory demands (encoding of $n = 5$ or $n = 10$ items; denoted $n5$ and $n10$, respectively) are somewhat more pronounced but still lie within the range of 1/10th of 1 Hz, as Fig. 5 indicates. Most important, the effects of memory demands are very consistent for different recording sites in both groups of subjects.

In contrast to these experimentally induced

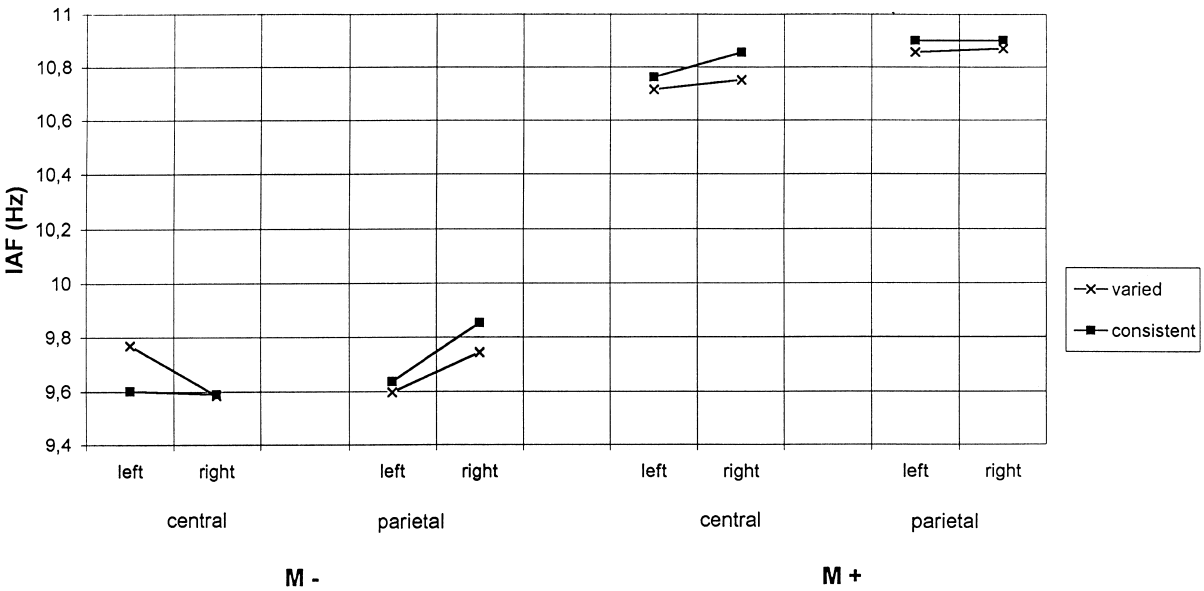


Fig. 4. At central and parietal leads increasing attentional demands lead to a decrease in alpha frequency as the results for good memory performance indicate. Although the variation of attentional demands shows up as a statistically significant interaction with other task variables, the magnitude of this effect is extremely small.

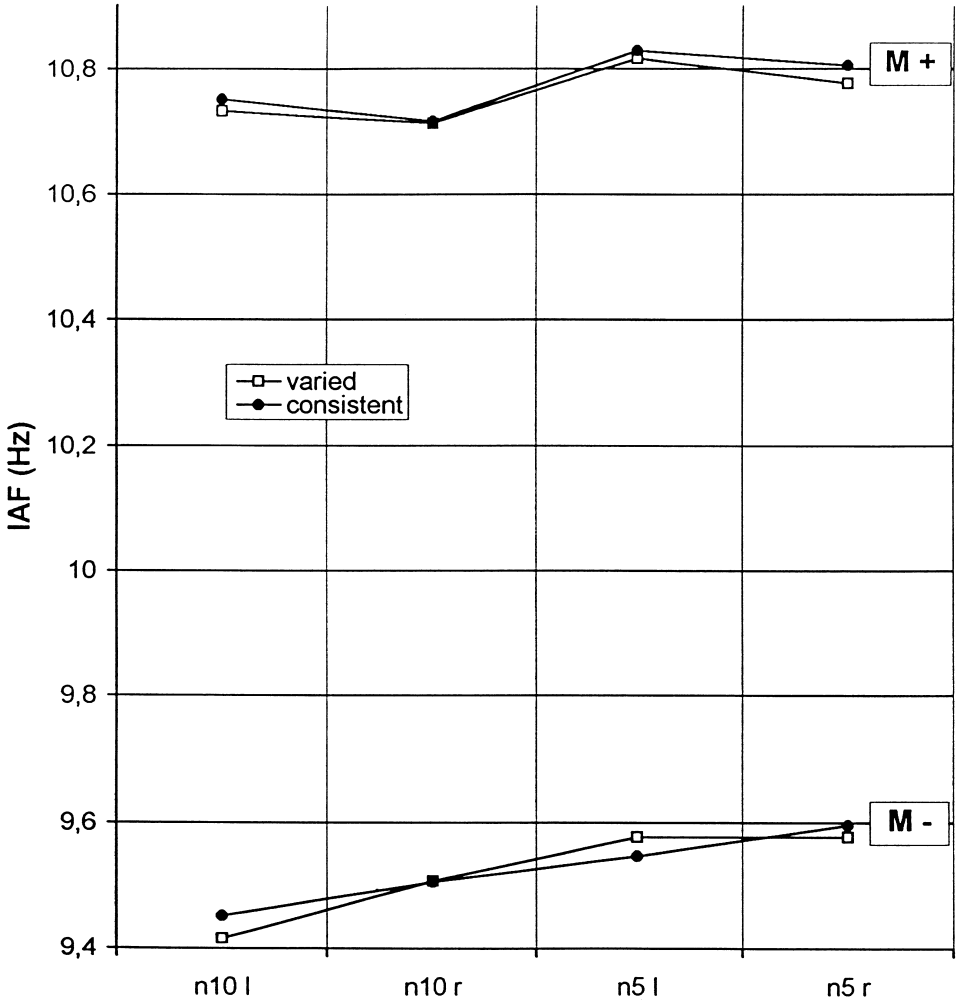


Fig. 5. High memory load (*n*10) tends to reduce alpha frequency in both groups of memory performers and at all recording sites of the left (l) and right (r) hemisphere.

variations, good and bad memory performers differ in the range of about 1 Hz. Thus, the interindividual differences in IAF which are related to differences in memory performance are more than 10 times larger than experimentally induced differences in the level of attentional and memory demands. The general pattern of results, thus, is that alpha frequency shows a tendency to decrease with increasing task demands. The by far strongest effect is a highly significant drop in IAF for bad memory performers of about 0.7 Hz with increasing experimental demands (see the drop in IAF for bad memory performers from the resting to the retrieval period in Fig. 3). This fact is particularly interesting because it demonstrates that the magnitude of event-related shifts in IAF may be much larger than the differences in IAF which were observed in response to variations in attentional and memory demands as mentioned above. This tonic shift in IAF for bad performers, which most likely reflects a general effect of increasing task difficulty does not blur but instead increases the group differences between good and bad performers. The group differences in IAF are replotted in Fig. 6 to show topographical differences over the time course of the experiment

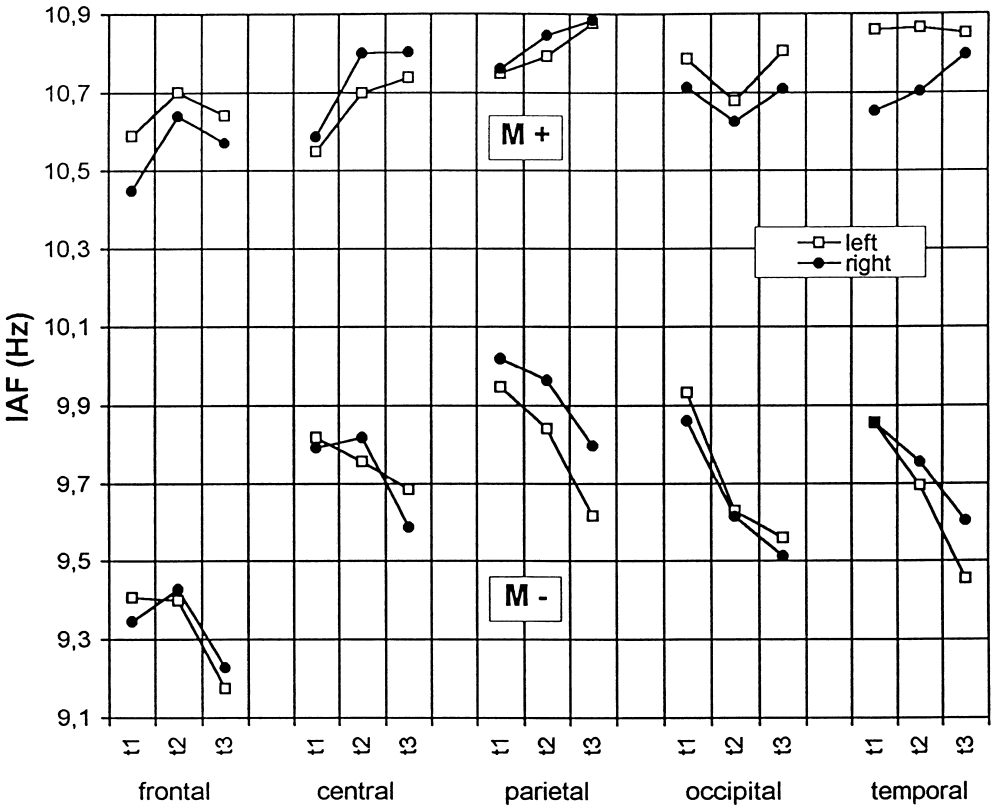


Fig. 6. Alpha frequency changes during the time course of an experimental trial. For good memory performers (M +), IAF tends to increase from the reference ($t1$) to the encoding ($t2$) and the retrieval ($t3$) interval at frontal, central and parietal recording sites of the left and right hemisphere. In contrast to good performers, bad memory performers (M −) exhibit a general tendency of a decrease in IAF. Note that during the encoding interval, a string of characters was visually presented for 2 s, whereas only one character was presented for 250 ms at the beginning of the retrieval interval. It is tempting to assume that the consistent drop in IAF at occipital recording sites during the encoding interval ($t2$) in both groups reflects visual encoding processes.

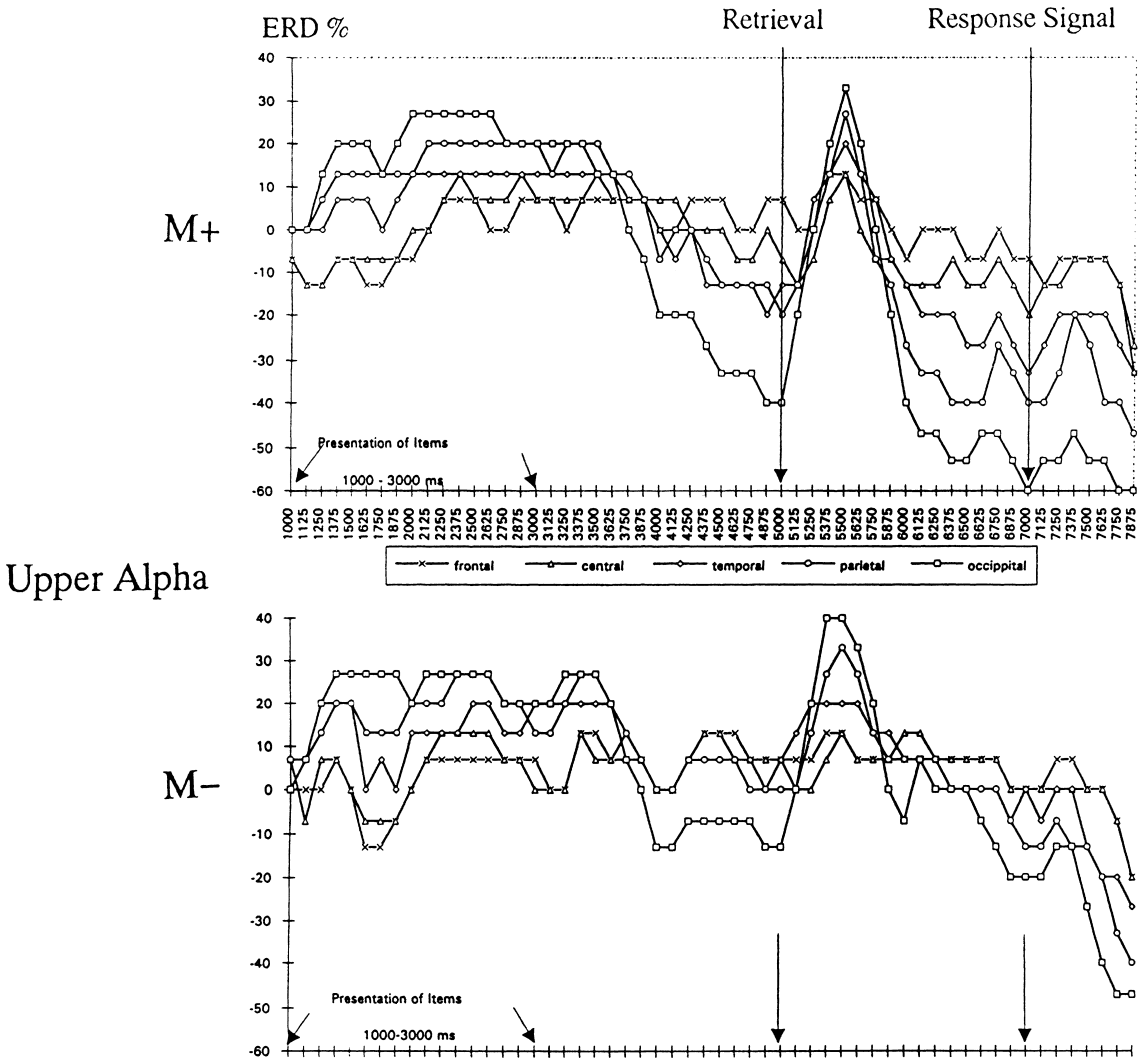
from the reference ($t1$) to the encoding ($t2$) and the retrieval ($t3$) interval.

Although for good performers, IAF stays at a high level during increasing task demands on the average (see also Fig. 3), even in this group a drop in frequency can be observed at occipital recording sites during the encoding interval. This drop, which is even more pronounced for bad performers, may reflect visual encoding processes. Comparing topographical differences between the two groups reveals that, in contrast to bad memory performers, good performers show a rather flat topographical distribution over the entire scalp and between different task demands (see the small differences between $t1$, $t2$, and $t3$). In contrast to good performers, bad performers have a particularly low alpha frequency at frontal recording sites. As can be seen by comparing IAF at frontal recording sites during the retrieval interval ($t3$), the difference in IAF between good and bad performers reaches a maximal value of about 1.5 Hz. Based on this result, it may be speculated that the low IAF of bad performers at frontal sites, which also can be observed during a pure resting period (Klimesch et al., 1993, Fig. 1), may reflect a lack to allocate attentional resources if task demands increase. Further evidence for the proposed interpretation comes from analyzing alpha power in the lower frequency band (see Fig. 7). These results indicate that bad

performers may have more limited attentional resources than good performers.

Finally, an interesting result is given by the fact that for bad performers the lowest frequency obtained throughout the entire experiment was found during retrieval and that IAF drops during retrieval at all recording sites over both hemispheres. This result is consistent with the hypothesis that retrieval is guided by widely distributed control processes. The fact that the topographically flat distribution of IAF for good performers is only slightly changed during retrieval also speaks against the view that retrieval processes are operating in particular cortical areas.

In summarizing the findings on IAF, the most important result is that for bad performers IAF generally decreases with increasing task demands. The most dramatic decrease in IAF of about 0.7 Hz was obtained in the retrieval period. A preliminary interpretation of these results is based on the general hypothesis that the speed of information processing is slowed down, once the capacity limits of the system are reached. Because



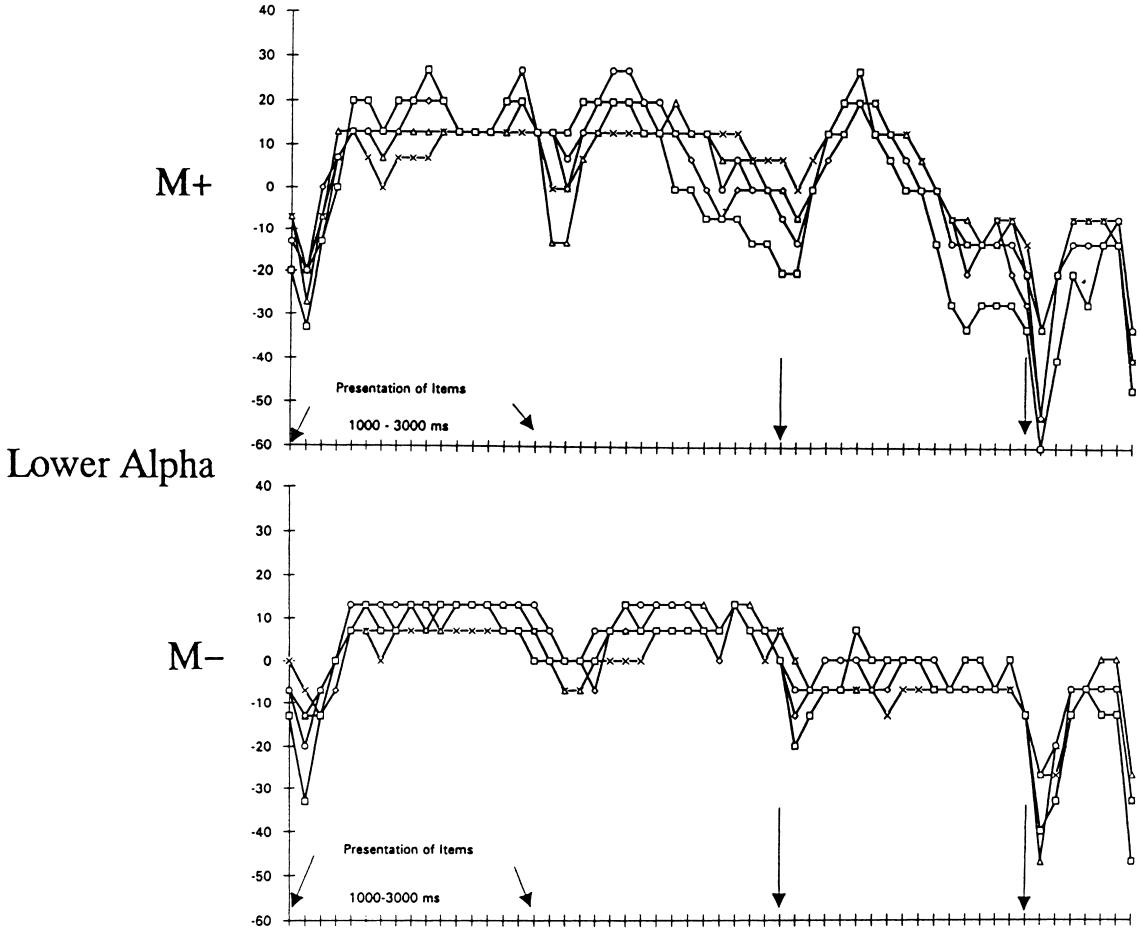


Fig. 7. Time course of ERD% in experiment three for the lower and upper alpha band during encoding and retrieval. In contrast to good memory performers, bad memory performers do not show any desynchronization in the lower alpha band during retrieval. In the upper alpha band, reflecting the retrieval of the presented character from memory, a pronounced desynchronization can be observed. In contrast to good performers, desynchronization lasts somewhat longer for bad performers. Note that the response signal (which invites a subject to give a verbal response) leads to a pronounced synchronization in the lower alpha band.

we have assumed that IAF reflects the speed of spreading activation and retrieval processes, we will expect that the speed of activating a cortical field (alpha field) and consequently alpha frequency serving this cortical field is slowed down. According to this view, alpha frequency should be correlated with the speed of memory processing. Behavioral evidence is in support of this view. As an example, in a reaction time experiment, Klimesch et al. (1988) have found that good memory performers are able to retrieve an item significantly faster from memory than bad performers.

The use of an individually adjusted alpha band is essential for observing the described relationships between task demands and alpha frequency, as control computations on our own data have revealed. Results reported in the literature are based on fixed frequency bands. Thus, it does not come as a surprise that in response to increasing task demands in some experiments an increase in alpha frequency was found (Earle, 1988, Fig. 2c) whereas in other experiments a decrease was reported (Zeller and Bente, 1983; Earle, 1988, Fig. 4). We already have emphasized that because of large interindividual differences, significant

portions of the alpha power distribution will fall outside a fixed frequency window. As an example, let us consider a subject with a low alpha frequency and let us assume that the lower alpha band falls below the frequency window of the fixed band which then covers only the upper alpha and some portions of lower beta. Consequently, a decrease or increase in frequency (i.e. a desynchronization or synchronization) of the lower alpha band cannot be detected in this case if a fixed band is used.

When interpreting IAF, we have to keep in mind that mean alpha frequency reflects the behavior of an entire population of different alpha frequencies. According to this view, a shift in IAF is not due to a frequency shift of a single alpha rhythm. Thus, an event-related frequency shift means that the power of some alpha frequencies may increase whereas others decrease. Physiologically, a power increase is interpreted in terms of synchronization (large cell populations respond with frequencies in a very narrow band) whereas a decrease in power is interpreted in terms of desynchronization (many but comparatively small cell populations respond with different frequencies). The question now is, how many frequencies there are and which frequencies of the entire population tend to change their power more than others. The first question is very difficult to answer, the second, is not. The results obtained from principal component analyses performed to define major spectral bands in the EEG have repeatedly shown that power values in the alpha band load on two different and orthogonal components (Mecklinger et al., 1992) with highest loadings in the lower and upper alpha band. As an example, Mecklinger et al. (1992, Fig. 2; Fig. 6), found two orthogonal components in the alpha band, one with highest loadings between 7 and 11 Hz and a second with highest loadings between 10 and 13 Hz. The overlap between the two components, thus, occurs between 10 and 11 Hz which is exactly that range where the peak frequency in a resting period (not reported by Mecklinger et al.) is to be expected for a sample of young subjects (Mecklingers subjects had a mean age of about 21 years). Taken together, these data indicate that power values of the lower and upper alpha band vary largely independent from each other and that resting alpha frequency marks (or is even created by) the overlap between the two alpha bands. Thus, instead of a single population of alpha frequencies, we have to deal with two different populations that operate largely independent from each other.

Reinterpreting our data on the basis of this evidence supports the idea that a decrease in IAF is due to a tendency towards desynchronization which is more pronounced in the upper than in the lower alpha band. Any attempt of a more precise interpretation, thus, depends on an understanding of the functional difference between the lower and upper alpha band. The experiments reported below provide evidence for the hypothesis that the lower alpha band reflects attentional processes whereas the upper alpha band reflects stimulus related cognitive processes. Attention in the sense of allocating processing resources probably reflects the monitoring function of thalamo-cortical feedback loops, whereas stimulus related cognitive processes are reflected by cortical oscillations in the upper alpha band. Thus, the transition from the lower to the upper alpha band may mark the transition of two networks, the network consisting of thalamo-cortical feedback loops and a cortical network. According to this hypothesis a decrease in IAF indicates strong cortical activation (because of a desynchronization of higher alpha frequencies) but comparatively slow working attentional processes (because of poor desynchronization of lower alpha frequencies). In trying to refine the preliminary interpretation stated earlier, we assume that the weak desynchronization of the lower alpha band leading to a decrease in IAF with increasing task difficulty reflects a slow read-out from cortical fields which, as a consequence, slows down information processing in the cortical storage network.

4. Shifts in alpha power and memory performance

Amplitude analyses in an individually defined lower and upper alpha band will be an important additional measure for studying cognitive processes and memory in particular. Topographical as well as interindividual differences in EEG

power are very large. Thus, an adequate way of measuring event-related shifts in EEG power should not be based on absolute but instead on relative differences as was first proposed by Pfurtscheller and Aranibar (1977) who have coined the term 'event-related desynchronization' or 'ERD'.

4.1. Event-related desynchronization (ERD) as a measure for shifts in alpha band power

ERD is defined as the percentage of a decrease or increase in band power during a test interval with respect to a reference interval. As an example, a single epoch (trial) in a memory experiment lasting several seconds may consist of the presentation of a warning signal (e.g. at the beginning of the 3rd s, i.e. at 3000 ms) and the presentation of a target or distractor (e.g. at the beginning of the 4th s, i.e. at 4000 ms). The reference interval typically is an interval of about 1 s which precedes the warning signal. Any interval, following the reference interval can be used as test interval. Thus, with a series of test intervals, the time course of shifts in band power can be monitored over the entire trial. The method of processing single epochs was described in detail elsewhere (Pfurtscheller and Klimesch, 1991) but for a basic understanding, the most important steps of the procedure will be briefly outlined below. First, artefact free EEG epochs are digitally band pass filtered within a defined frequency band. Second, in order to get a simple measure of band power, each sample point (i.e. band pass filtered EEG amplitude) of each epoch is squared. Third, these data are averaged separately over epochs and electrodes and after selecting the reference interval, the remaining part of the averaged epoch is segmented in a series of test intervals. Time resolution is limited by the length of the test intervals which can be as small as 125 ms. As an example, if sampling frequency is 128 Hz and test intervals have a length of 125 ms, each test interval represents the average of 16 data points. Based on these computations, the percentage of an increase or decrease in band power is calculated for each electrode and experimental condition according to the following simple equation:

$$ERD\% = ((\text{band power reference interval} - \text{band power test interval}) \div (\text{band power reference interval})) \times 100$$

It is important to note that a positive value indicates a decrease in band power whereas a negative value indicates an increase in band power with respect to the reference interval. Thus, positive values reflect a state of desynchronization, negative values a state of synchronization or ERS (event-related synchronization, see Pfurtscheller, 1992).

The most basic finding for the upper alpha band is that those cortical areas which are involved in the processing of a task tend to desynchronize, whereas other areas which most likely are not primarily involved either tend to synchronize or show no power changes at all. Examples of ERD maps showing the encoding of a visually presented word and the preparation for a verbal response can be found, e.g. in Pfurtscheller and Klimesch (1992) (Fig. 7).

4.2. ERD and memory processes

When analyzing the data of Experiment 3 on the basis of event-related power changes in the individually determined lower and upper alpha band, an interesting difference between good and bad memory performers can be observed. Whereas good and bad memory performers show a pronounced desynchronization in the upper alpha band, bad performers lack any significant desynchronization in the lower alpha band during retrieval. A similar but somewhat weaker tendency was also found for the encoding interval. Comparing ERD and IAF reveals a striking similarity between the two measures of alpha activity. As was reported above, in contrast to good memory performers, bad performers show a lower IAF which drops even further if task demands increase. A drop in IAF is due to a shift in alpha power towards lower frequencies which reflects a strong asymmetry in the amount of desynchronization in the lower and upper alpha band. This

exactly is what the ERD analysis shows: In the lower alpha band, bad performers show either no or a much smaller ERD than good performers (see Fig. 7). The conclusion, thus, is that bad memory performance is related to a weak or insufficient desynchronization of the lower alpha band. But what is the meaning of this result?

4.3. Attention and memory: the functional meaning of the lower and upper alpha band

When trying to interpret the differences in the lower alpha band for good and bad memory performers, it is necessary to focus on the functional meaning of the two alpha bands. A first hint can already be found in Fig. 7. Inspecting the time course of ERD reveals that the effect of the response signal is a strong short-lasting synchronization which appears in the lower alpha band only. Subjects were instructed to wait with their response until the response signal appeared 2 s after a target or distractor was presented. The short-lasting synchronization in the lower alpha band indicates the occurence of an expected event on which the subject gives an overt response to a decision he or she already knows. Thus, the effect of the response signal in this experiment obviously is to induce a shortlasting relaxation which is correlated with a decrease in attention.

With respect to the use of the concept 'attention', some clarifying remarks will be useful. Psychological theories of attention usually focus on two different questions (Treisman, 1986). One refers to those mechanisms underlying the allocation of limited processing resources, the other to selective attention. These two aspects are also stressed by Posner (Posner and Boies, 1971) who distinguishes between three components of attention: alertness, selectivity, and processing capacity (for an extensive review see Posner, 1975). Alertness — a concept closely related to arousal, which Posner avoids because of its emotional connotation — is further subdivided into phasic and tonic alertness. Phasic changes in alertness are under volitional control and occur at a rapid rate, whereas tonic changes are not (or less) under volitional control and occur at a much slower rate. A typical example of a phasic change is the increase in alertness after the presentation of a warning signal. Tonic changes, on the other hand, occur as a function of fatigue, distress, diurnal rhythms and changes over the life cycle. According to Posner, alerting stimuli such as a warning signal have no selective effects on information processing. Selective effects are due to 'pathway activation' which serves to activate particular memory traces. According to this view, selective attention and phasic alertness are essential requirements for any cognitive process. A memory search as an example of a cognitive process will be guided by selective attention, whereas phasic alertness is a comparatively distributed mechanism which allocates — or prepares to allocate — processing resources.

In Klimesch et al. (1992), the effects of phasic alertness on the two alpha bands were studied. In one of the two experiments, subjects were required simply to read a series of words and numbers. The only experimental manipulation was the variation of the time period between the presentation of the warning signal and the presentation of a word or a number. According to this manipulation, the warning signals (denoted W1, W2, W3) appeared randomly either 1, 2 or 3 s before the imperative stimulus. The results are depicted in Fig. 8 and demonstrate that the effect of the warning signal shows up in the lower alpha band only. These findings were replicated in yet another experiment in which the additional effects of expectancy were investigated (Klimesch et al., 1992).

In contrast to the response signal in Klimesch et al. (1993) which reminded the subject to respond, an alerting effect can be observed in Fig. 8. The alerting effect is documented by a phasic desynchronization of the lower alpha following the presentation of a randomly spaced warning signal (W1, W2 and W3). Comparing the time course of desynchronization in the two alpha bands reveals that the largest change in alpha power was found for the lower alpha band in response to the imperative stimulus. This result, which was replicated in Experiment 2 of Klimesch et al. (1992) indicates that although a warning signal makes a subject more alert, the strongest increase in alertness — most likely reflecting the

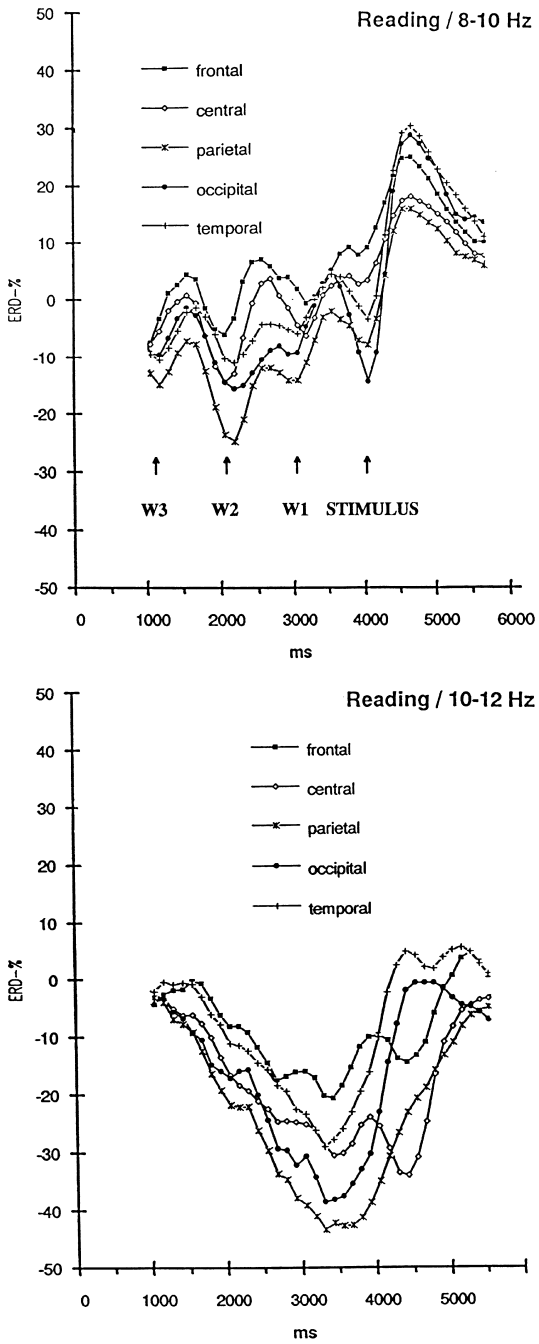


Fig. 8. ERD% for a warned stimulus presentation of a word or number. The subject's task simply was to read the presented stimulus. The warning signal appeared randomly either 3, 2 or 1 s before the stimulus was presented. Note that the effects of the warning signal show up in the lower alpha band only.

allocation of processing resources — is associated with the encoding of the imperative stimulus. In the upper alpha band, the time course of desynchronization shows a strikingly different pattern. Here, desynchronization occurs in response to the encoding of the imperative stimulus, whereas the effects of the warning signals are completely lacking. Because of the divergent results found for the two alpha bands, it seems plausible to assume that the upper alpha band responds selectively to the encoding of the stimulus, whereas the lower band reflects attentional processes. This interpretation is in good agreement with the fact that, compared to the lower band, the increase in ERD is much more localized in the upper band with maximal desynchronization at temporal and occipital recording sites (see Fig. 8). Because of the visual-verbal nature of the task in both experiments of Klimesch et al. (1992), these brain areas would have been expected to exhibit the most significant desynchronization, i.e. increase in ERD.

The suggested hypothesis concerning the functional difference of the two alpha bands was tested in an experiment which analyzed the Dm-effect by means of ERD (Klimesch et al., 1996). The Dm-effect (or simply 'Dm' for Differences based on later memory performance; Paller et al., 1987) refers to the result that during the encoding of pictures or words that were remembered in a later recall or recognition test, some components of the event-related brain potential (ERP) are significantly more pronounced (more positive going) than for items which cannot be remembered (see also section 3.4. below). According to the proposed hypothesis regarding the functional meaning of the lower and upper alpha band the following prediction can be made. If the Dm-effect is related to differences in attention, the remembered words should exhibit a significantly stronger desynchronization in the lower alpha band. However, if the Dm-effect reflects differences in cognitive processes (such as different encoding strategies), the remembered words should show a pronounced desynchronization in the upper alpha band. The results of Klimesch et

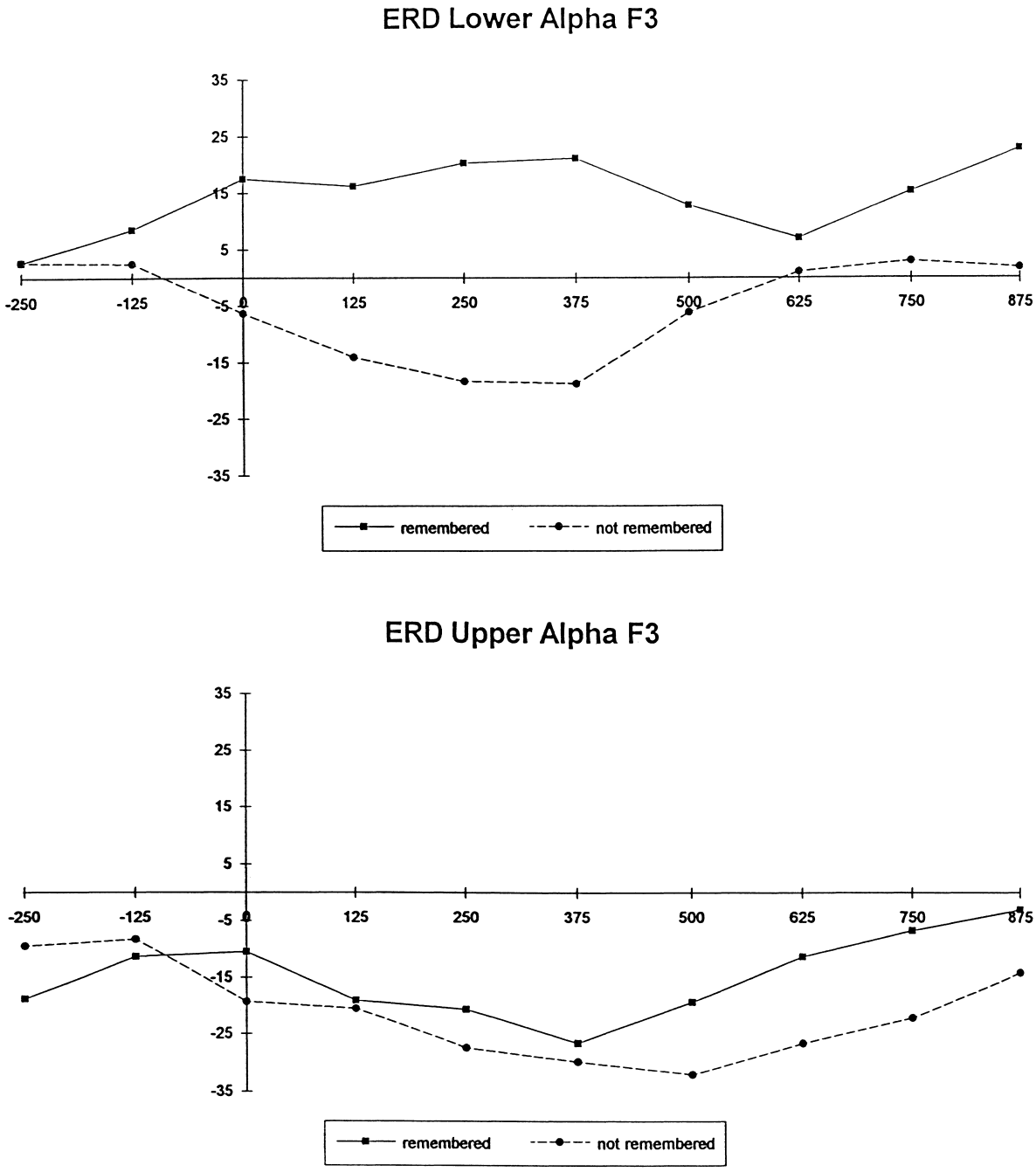


Fig. 9. ERD% in an incidental memory paradigm. Subjects were presented words, but were not informed that their memory performance will be tested later. The strongest Dm-effect was found in the lower alpha band and at *F*3. The topography and the fact that the Dm-effect arises already before the stimulus is presented, provides additional evidence for the hypothesis that the Dm-effect is related to fluctuations in attention.

al. (1996), which are depicted in Fig. 9, support the idea that the Dm-effect is related to attentional processes because the Dm-effect shows up in the lower but not in the upper alpha band.

4.4. The upper alpha band and semantic memory

Finally, let us consider the hypothesis proposed in Klimesch et al. (1996) that processes in semantic LTM are primarily reflected by EEG frequencies in the upper alpha band (or higher frequencies such as beta or gamma which are not consideredhere), whereas episodic encoding processes in working memory are reflected in a different frequency band, possibly in the theta band (for a discussion of this issue see also Klimesch et al., 1994). Because we have found some evidence that the lower alpha band reflects attentional processes, we have to assume that semantic processes show up in the upper but not in the lower alpha band. In the experiments reported so far, semantic as well as episodic memory processes played an important role. Thus, the critical test whether or not the alpha band selectively reflects semantic memory processes, had still to be carried out in a further study (Klimesch et al., 1994). In this study, a design was used that is described in Experiment 4 by Kroll and Klimesch (1992) and that already prooved useful to distinguish semantic from episodic retrieval processes on the basis of behavioral measures (RTs). The experimental design consisted of two parts. Subjects first performed a semantic congruency task in which they had to judge whether or not the sequentially presented words of concept-feature pairs (such as 'eagle-claws' or 'pea-huge') are semantically congruent. Then, without prior warning, they were asked to perform an episodic recognition task. This was done in an attempt to prevent subjects from using semantic encoding strategies and thus to increase episodic memory demands. In the episodic task, the same concept-feature pairs were presented together with new distractors (generated by repairing known concept-feature pairs). Now subjects had to judge whether or not a particular concept-feature pair was already presented during the semantic task. The results of Kroll and Klimesch (1992) indicated that semantic features speeded up semantic but slowed down episodic decision times (for an extensive review on this topic see also Klimesch, 1994). With respect to the purpose of the present study, this result indicates that semantic and episodic memory processes can well be differentiated by using the design underlying Experiment 4 in Kroll and Klimesch (1992). According to the proposed hypothesis, it is expected that only in the semantic task the most pronounced desynchronization should be observed in the upper alpha band. Because pairs of items are presented and a subject can perform the episodic and semantic task only after the second item of a pair (i.e. the feature) is presented, the increase in upper alpha ERD as a response to increasing semantic task demands is expected only for the time period following the presentation of the feature. It is important to keep in mind that alpha power is well known to decrease with increasing task difficulty. From the results found in Kroll and Klimesch (1992) we know that the episodic task is much more difficult than the semantic task. Thus, if task difficulty would be the only factor which is reflected by an event-related decrease in alpha power (i.e. an increase in ERD) we would expect the most pronounced increase in ERD to be observed during the presentation of the feature in the episodic task. This, however, is not the case, as the results depicted in Fig. 10 show. In summarizing, the following three results are of interest: First, upper alpha ERD increases during the presentation of the feature in the semantic task. Second, lower alpha ERD increases in the more difficult episodic task. Third, theta ERD decreases (i.e. theta power increases) in the episodic task.

As Fig. 10 indicates, task demands lead to a desynchronization in the upper and lower alpha band (i.e. to a decrease in band power) but to a synchronization in the theta band (i.e. to an increase in band power). This fact seems to indicate that different neural generators are involved (see, e.g. the review in Lopes da Silva, 1992). The transition from theta synchronization to alpha desynchronization can be seen most clearly if ERD is analyzed in 1-Hz bands (Schimke and Klimesch, 1996). Fig. 11 shows the results for the

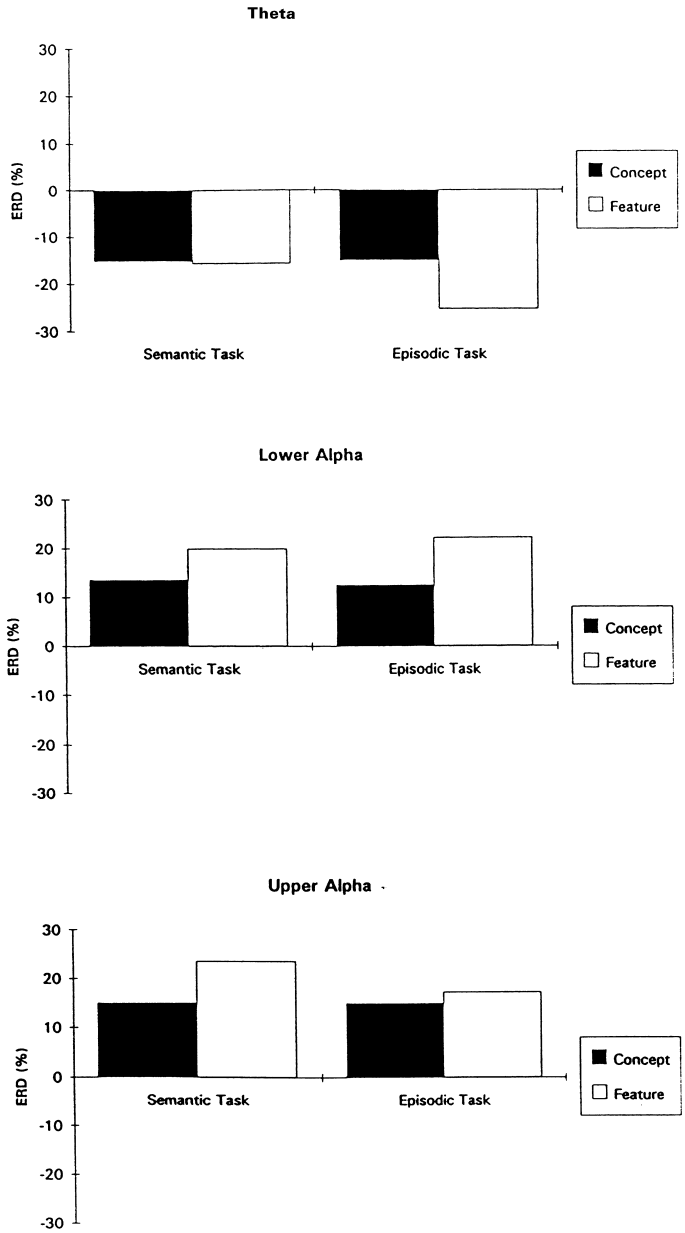


Fig. 10. In a semantic task, subjects had to judge whether a feature (such as 'green') is consistent with a concept (such as 'canary'). In an episodic task, subjects had to indicate whether a particular feature was already paired with a particular concept. Because the episodic task is much more difficult than the semantic task, the amount of alpha desynchronization should be much more pronounced in the episodic as compared to the semantic task. This, however, is not the case. Note that both tasks can be performed only after the feature is presented.

encoding of the feature in the semantic task (see with this the left part of Fig. 10). It also is important to emphasize that these results too are based on individually determined alpha bands. Consequently, the lower frequency limit of the lower alpha band is the upper frequency limit of the theta band. In other words, the theta band was aligned to the individual frequency of the

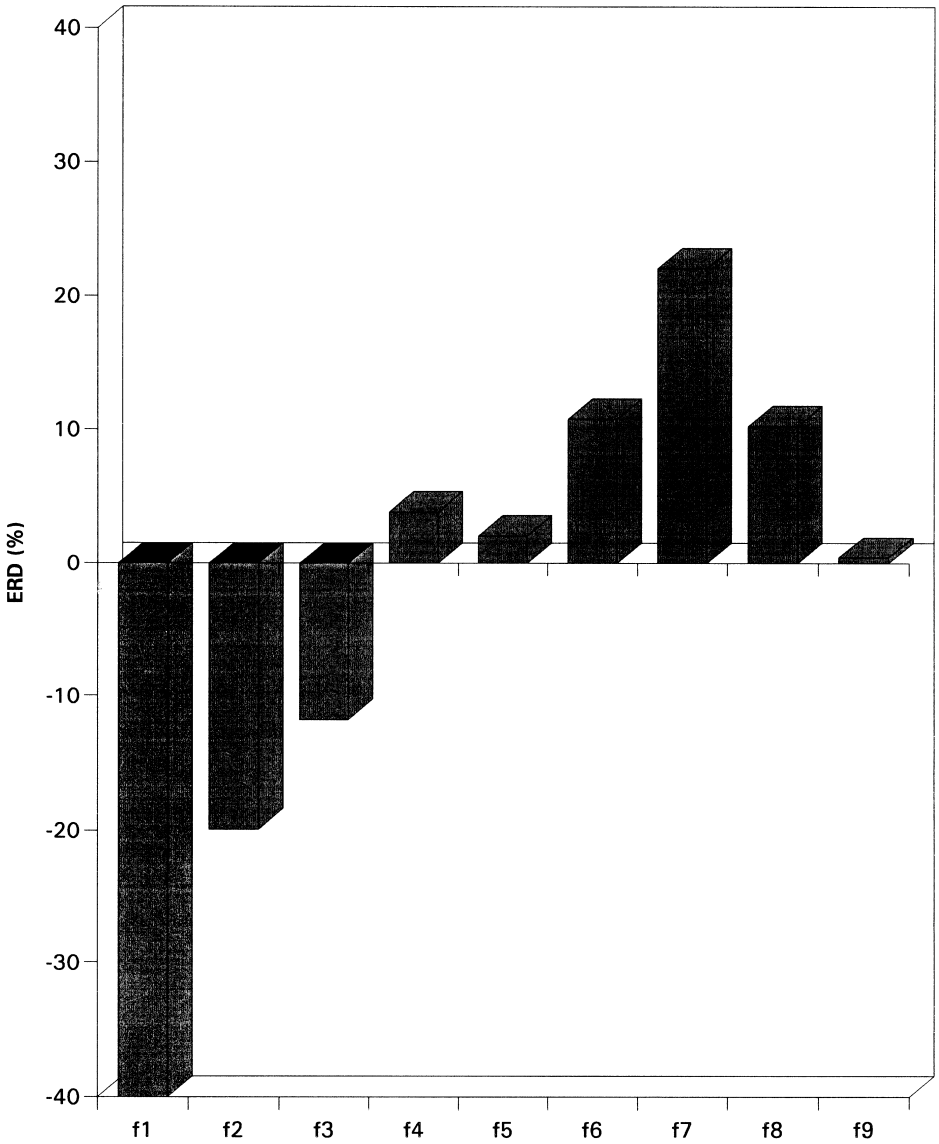


Fig. 11. When analyzed in 1-Hz bands (Schimke and Klimesch, 1996), the differential effects of synchronization in the theta bands ($f1$–$f4$) and desynchronization in the alpha bands ($f5$–$f9$) become quite pronounced. The frequency analysis is calculated in steps of 1 Hz bands below and above the individually determined alpha gravity frequency. Averaged over the sample of subjects, theta lies in the range of 3.4–7.4 Hz and alpha within 7.4–12.4. Accordingly, $f7$, the frequency band with the strongest desynchronization, represents the frequency range of the alpha peak.

alpha band. Control computations have shown that the reported results cannot be replicated with fixed frequency bands in the alpha and theta band. This fact supports the idea that different frequency domains such as theta and alpha do not vary independently but instead are adjusted to each other.

5. Discussion

The most general conclusion is that alpha frequency is a sensitive measure for cognitive performance or cognitive processing capacity. Alpha frequency varies inter- and intraindividually as a

function of processing capacity. The decrease in alpha frequency with increasing age and the lowered alpha frequency as a result of a variety of neurological disorders are well-known examples underlining the validity of a general relationship between alpha activity and cognitive performance.

Memory and attentional resources are the most important factors determining cognitive processing capacity. The specific relationship between alpha frequency and memory performance is demonstrated by the fact that alpha frequency is significantly and positively correlated with memory performance (e.g. Klimesch et al., 1990). In age matched samples, the mean difference in alpha frequency between good and bad memory performers is about 1 Hz (see Fig. 3). Because in normal age matched subjects the standard deviation of alpha frequency equals 1 Hz (see Fig. 2), differences in memory performance can account for a good deal of this normal variability of alpha frequency.

It is important to emphasize that even in a resting period alpha frequency differs between good and bad memory performers. However, with increasing task demands and memory demands in particular, the difference in alpha frequency increases between the two groups. Whereas good memory performers show a tendency of a slight increase in alpha frequency at anterior recording sites, bad memory performers show a pronounced decrease over all recording sites. A possible interpretation of this finding may be based on the idea that within the capacity limits of the cognitive or memory system in particular, increasing task demands lead to an increase in frequency. If, however, the capacity limits are reached, any further increase in task demands leads to a decrease in alpha frequency. Results supporting this interpretation are: (1) the fact that alpha frequency for bad performers (who by definition have a lower memory capacity than good performers) decreases at all recording sites; (2) the fact that under the condition of high memory load (encoding of 10 characters), which exceeds the capacity of short-term memory, alpha frequency decreases for bad as well as for good performers; and (3) the fact that during the encoding interval (t_2), where subjects had to use all their cognitive capacity to encode and rehearse the visually presented characters, occipital recording sites show a further drop in alpha frequency in both groups. Finally, the finding that the experimental variation of memory load (Fig. 5) showed a much more systematic effect than the variation of attentional demands (Fig. 4) also underlines the specific role of alpha frequency for memory processes.

In terms of alpha power desynchronization, a decrease in alpha frequency indicates that the upper alpha band shows a more pronounced desynchronization than the lower alpha band. Therefore, it is to be expected that bad performers (with their pronounced drop in alpha frequency) show only a weak desynchronization in the lower alpha band. The results depicted in Fig. 7 support this interpretation and show that during the critical retrieval interval, bad performers lack any significant desynchronization. Because other studies (Klimesch et al., 1992) have indicated that the lower alpha band reflects attentional processes (Fig. 8), it may be concluded that bad memory performers have difficulties in allocating their memory processing resources. This interpretation is in agreement with the finding that bad performers have a particularly low alpha frequency at anterior recording sites and that desynchronization in the lower alpha band is related to the Dm-effect (Fig. 9) which may reflect differences in attention during encoding.

Comparing alpha frequency and alpha desynchronization in the two frequency bands leads to the following fundamental question: Is a frequency shift 'real' or is it caused by an asymmetric desynchronization of the two alpha bands? Or in other words, is the lower alpha frequency of bad performers due to a chronic lack of desynchronization in the lower alpha band and thus to a chronic attentional deficit or is it an indicator of a general slower speed of information processing? In trying to answer this question, let us first consider alpha resting frequency which reflects a state of synchronization. It is well established that the speed of information processing decreases with age, just as alpha frequency does. Because good and bad performers differ even in a resting period, it seems plausible to assume that this frequency difference is 'real' in the sense that bad

performers have a general slower speed of information processing. However, the fact that with increasing task demands a further drop in alpha frequency can be observed speaks for an asymmetric desynchronization, which in this case means that primarily the upper, but not the lower alpha band desynchronizes. Thus, during rest, frequency differences may be 'real' differences, but event- or task-related frequency shifts may not be real but instead reflect an asymmetry in the amount of desynchronization in the two alpha bands.

In terms of the theoretical framework outlined earlier (Fig. 1) memory processes may be reflected by a spread of alpha frequency in the cortical storage network, whereas attention as a subcategory of monitoring processes may be related to the activity of the thalamo–cortical monitoring network. Thus, the upper alpha band should be related to neural activity in the cortical storage network, whereas the lower alpha band may be related to neural activity stemming from the thalamo–cortical network. Monitoring processes serve to allocate processing resources in the sense that they guide search and retrieval processes. Therefore, monitoring processes are closely related to attention. A lack of desynchronization in the thalamo–cortical network reflects inefficient search and retrieval processes and, thus, poor memory performance. Of course, the proposed interpretation is still highly speculative, but may be useful in planning new experiments that focus on shifts in alpha activity as a function of different types of attentional and memory demands.

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